

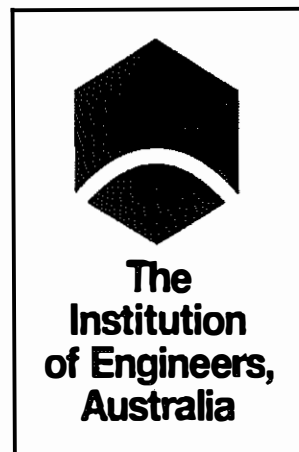
August 1998 Wollongong Storms

WOLLONGONG UNIVERSITY, WOLLONGONG

Chairperson: Michael Boyd

Organising Committee: Mark Babister
Michael Boyd
Jim Irish
Scott Little

23rd FEBRUARY 1999



Sydney Division
Water Engineering Panel

August 1998 Wollongong Storm

University of Wollongong, Tuesday 23 February 1999

- 9.00 Registration**
- 9.15 Welcome and Introduction**
- 9.25 Wollongong Flash Flood Event**
Julie Evans & Ben Bewick
Bureau of Meteorology
- 9.50 State Emergency Service Response**
Stephen Oppen
State Emergency Service
- 10.20 Morning Tea**
- 10.40 Social Impacts**
Peter Olney
DOCS
- 11.05 Disparate Understanding of Hydraulic Evidence**
Jim Irish
University of Technology, Sydney
- 11.30 Insurance Emergency Scheme**
John Cooper
Insurance Emergency Scheme
- 11.55 Influence of the Illawarra Escarpment on Long Duration Design
Rainfalls and the Implications for Floodplain Management**
Scott Little and Mark Babister
Webb McKeown & Associates
- 12.20 Lunch**
- 1.20 Workshop & Discussion**
- 2.20 Close**

**THE INSTITUTION OF ENGINEERS, AUSTRALIA
SYDNEY DIVISON
WATER ENGINEERING PANEL
AUGUST 1998 WOLLONGONG STORMS**

ORGANISATION

Auburn Council
Bureau of Meteorology
Bureau of Meteorology
Bureau of Meteorology
Camden Council
Cooper Adjusting P/L
Dept of Urban Services, Act Roads & Stormwater
Dept of Urban Services, Act Roads & Stormwater
Dept of Urban Services, Act Roads & Stormwater
Dept of Urban Services, Act Roads & Stormwater
Dept of Urban Services, Act Roads & Stormwater
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DLWC
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Ecowise
Environmental Act
Environmental Management
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Fairfield City Council
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Forbes Rigby
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Gilbert Rose Consulting P/L
John M Daly & Associates P/L
Jones Nicholson P/L
Jones Nicholson P/L
K G Macoun & Associates P/L
Kinhill P/L
Kinhill P/L
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NAME

Andrew Mcphail
Anita Fletcher
Phillip Allen
George Stamatakos
Julie Evans
Gordon Mckay
Hugh Bruist
Andrew Celona
John Cooper
Hugh Crawley
Ian Pickering
Karl Cloos
Steve Bresnan
Kirilly Dickson
James Erskine
Bruce Green
David Avery
Janice Green *Not Here*
Garry Hopkins
Norm Mueller
Chris Ryan
Barry Le Plastrier
Michelle Maher
Robert French
Terry Lustig
Margie Azmi
Andy Chin
John Price
Jacki Low
Steve Roso
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Terry Hams
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Andrew Macleod
Robert Dinard
Wendy Cox

Lawson & Treavor
Lawson & Treavor
Lyal & Macoun Consulting Engineers
Lyal & Macoun Consulting Engineers
Lyal & Macoun Consulting Engineers
Mbk South Coast P/L
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Natural Hazards Research Centre, Macquarie Uni
Natural Hazards Research Centre, Macquarie Uni
Nrma
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Patterson Britton & Partners
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Rammy Associates
SES
Shoalhaven City Council
Strathfield Council
Sydney Water Corporation
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Sydney Water Corporation
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Water Research Laboratory, Unsw
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Webb McKeown & Associates
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Wingecarribee Shire Council
Wingecarribee Shire Council
Wingecarribee Shire Council
Wollongong City Council

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Ajith Goonatilleke
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Michael Brearley
Peter Nunn
Carey McIntyre

“Wollongong Flash Flood Event”

Julie Evans & Ben Bewick

Bureau of Meteorology

The Wollongong Flash Flood Event

17 August 1998

Julie Evans and Ben Bewick
NSW Regional Office
Bureau of Meteorology

1. Introduction

During the evening of 17 August 1998 between 1700 and 2000 Eastern Standard Time (EST) extremely heavy rainfall near the Illawarra escarpment to the northwest of Wollongong caused flash flooding, extensive damage to property and one fatality. The rainfall during this three hour period was the most intense part of a 96 hour rainfall event extending from 1800 EST Saturday 15 August to 1800 EST Wednesday 19 August 1998. Prior to the heaviest rainfall, between 1700 and 2000 on 17 August, parts of the Wollongong catchment had already received up to 375 mm of rain during the previous 47 hours. This paper is a synopsis of a report currently being prepared by the Bureau of Meteorology.

The highest rainfall total recorded over the entire 5-day period was 745 mm at Mt Ousley, while the highest 24 hour total to 9am on Tuesday 18 August, also at Mt Ousley, was 445 mm. Analysis of the rainfall intensity for durations from 5 minutes to 72 hours has revealed return periods over 1 in 100 years for durations between 30 minutes and 6 hours along the top of the escarpment near Mt Ousley, Rixons Pass, Bulli Pass, Balgownie Reservoir and Beth Salem.

Many intense rainfall events in New South Wales have been located along the coastal escarpment of the Great Dividing Range. Near Wollongong, the terrain rises steeply to 400m about 4 km from the coast rendering the region susceptible to heavy rainfall events. Shepherd and Colquhoun (1985) and Nanson and Hean (1984) have described the most extreme event recorded in the area, which occurred near Dapto (12 km southwest Wollongong) in February 1984. The maximum 24-hour total of 796 mm was recorded at Wongawilli. Davison (1981) has surveyed flood events over the period 1945-1977 based mainly on newspaper reports when daily rainfall at Mt. Keira was greater than 50mm. Steep terrain has also played a role in flooding at Coffs Harbour. On November 23 1996, just under 400mm of rain fell on the nearby hills in 4 hours resulting in a flash flood which generated insurance losses of \$30 million (Speer and Leslie, 1998).

Speer and Geerts (1994) have studied the climatology of flash-floods in the Sydney metropolitan area, just to the north of Wollongong, from 1957 to 1990 identifying four main types of synoptic flow conducive to heavy rainfall: easterly troughs, lows, pre-frontal flow and post-frontal flow. All types are associated with upper-level low pressure troughs or deep cut-off low pressure systems.

2. Synoptic Situation.

On 17 August 1998 at 1200 EST a low pressure system in the northern Tasman Sea and a high pressure system in the southern Tasman Sea, were directing a moist easterly airstream towards the New South Wales coast (Fig. 1). In addition, an inland trough extended through Queensland, New South Wales and into Victoria and a mesoscale trough was located just offshore from Sydney and the southern New South Wales coast (not shown in Fig.1). At 1500 EST, a mesoscale low was analysed within this trough to the south of Sydney. By 1800 EST this system had moved onshore (Fig 2), remaining inland for the next nine hours, returning to the coast around 0300 EST Tuesday 18 August.

At 0900 EST 17 August a sharp upper trough at 500 hPa was oriented from the northeast to the southwest over inland New South Wales (Fig 3). This upper trough moved slowly to the southeast during the day, to be located near a line from Newcastle to Wagga Wagga at 1500 EST. The trough was over the Wollongong area at about 0300 EST Tuesday 18 August, moving offshore by 0900 EST.

The rainfall was enhanced where the vertical motion of the moist onshore airstream was maximised. This occurred within the mesoscale trough/low where the upward motion was increased by the convergence of the low level wind into the system. The steep topography also caused additional convergence and uplift of the airflow. As the upper trough approached the Wollongong area, the air immediately to the southeast of the trough axis would have undergone broadscale lifting and destabilisation, further enhancing vertical motion and associated heavy rainfall.

3. Rainfall Analysis

3.1 Spatial Distribution

Rainfall isohyets for the 5-day period ending 0900 EST Thursday 20 August are shown in Fig. 4. These isohyets are based on available pluviograph and raingauge data. Data from unofficial sources is not included. There are few stations west of the Illawarra escarpment. The maximum point rainfall was recorded at Mt. Ousley (745 mm) followed by Beth Salem (712 mm), Rixons Pass (672mm) and Bulli Pass (654mm). The area with recorded totals exceeding 600mm was centred along the top of the escarpment from Bulli Pass to Mount Ousley extended southwestward to Dombarton Loop and the Avon and Cordeaux Dams.

Figure 5 shows the 24 hour totals to 0900 on Tuesday 18 August near the storm centre, again using data from both raingauge and pluviometer networks. The restricted 24-hour totals (to 0900) over 400mm were recorded at Mt. Ousley (445 mm), Rixons Pass (440 mm), Wonoona Heights (437 mm) and Bulli Pass (411mm). The maximum unrestricted 24-hour intensities, obtained from pluviometer data that are not limited to one 24-hour measurement taken at 9 am, were greater and are discussed below.

3.2 Temporal Patterns

The temporal patterns of accumulated rainfall for pluviometer stations near the storm centre during the period Saturday 15 August to Wednesday 19 August are shown in Figure 6.

The heaviest rain fell in the 13 hours between 0800 and 2100 on Monday 17 August, when, at the centre of the storm, over 50% of the storm total was recorded. More than half of that amount occurred between 1700 and 2000. However, significant rainfall had already fallen prior to 0800 on Monday.

The first rainfall period commenced about 1800 on Saturday 15 August and continued until 11 am on Sunday 16th. During this period rainfall totals over 100mm were recorded at many stations. Rainfall recommenced around 0100 Monday morning, but near the escarpment, falls remained relatively light until 1700 with the exception of a 30 minute period between 1500 and 1530.

From 9 am 15 August to 1700 EST on 17 August, the beginning of the most intense 3-hour period of rain, rainfall totals of the order of 200 – 375 mm had fallen over parts of the catchment. Rainfall totals between 1700 and 2000 EST on 17 August were the highest along the eastern edge of the escarpment and slightly to the west near Beth Salem.

3.3 Maximum Intensities and return periods

The most intense period of rainfall was centred around 1900 on 17 August. The highest intensities were recorded at Rixons Pass, Mt. Ousley, Balgownie Reservoir, Bulli Pass and Beth Salem. Average Recurrence Intervals (ARIs) exceeded 100 years at one or more of these five stations for durations between 30 minutes and 6 hours, and at the first four stations for a duration of three hours. The rainfall intensities at Rixons Pass for durations from 6 minutes to 72 hours are shown in Figure 7.

Eastward from the escarpment the rainfall return periods quickly decreased. For example the maximum ARIs at Russell Vale, Thirroul and Keiraville were between 20 to 50 years and near the coast at Wollongong City they had dropped to between 2 to 5 years.

Acknowledgements

Pluviometer data was provided by the Manly Hydraulics Laboratory, Sydney Water, Wollongong City Council and the Roads and Traffic Authority. The ARI information was provided by the Hydrometeorological Advisory Service of the Bureau of Meteorology.

References

Armstrong, J.G. and Colquhoun, J.R. 1976. Intense rainfall from the thunderstorms over the Sydney Metropolitan and Illawarra district – 10 and 11 March 1975. *Tech. Rep. 19*, Bur. Met., Australia.

Davidson, A.R. 1981. Flooding in Wollongong 1945-1977: A survey. Occasional Pap. 30, Faculty of Military Studies, Dept of Geography, Univ of NSW.

Nansen, G.C. and Hean, D. 1985. The West Dapto flood of February 1984: Rainfall Characteristics and Channel Changes. *Australian Geographer*, 16, 249-258.

Shepherd D.J. and Colquhoun, J.R. 1985. Meteorological aspects of an extraordinary flash flood event near Dapto, New South Wales. *Aust. Met. Mag.*, 33, 87-102.

Speer, M and Geerts, B. 1994. A synoptic-mesoalpha-scale climatology of flash-floods in the Sydney metropolitan area. *Aust. Met. Mag.*, 43, 87 – 103.

Speer, M.S. and Leslie, L.M. 1998. Quantitative precipitation forecasting for an extreme mesoscale flash flooding event. 16th Conference on Weather Analysis and Forecasting, Phoenix, American Meteorological Society. p 144.

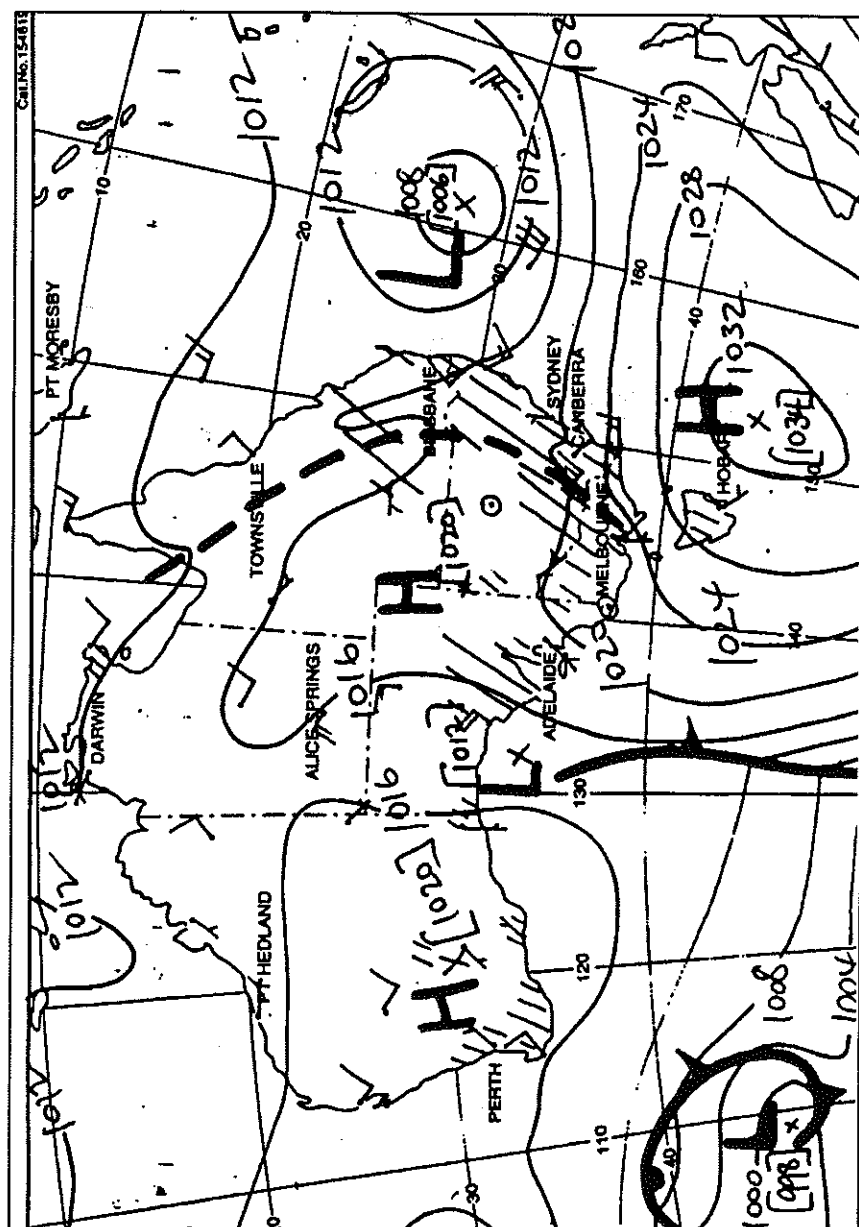


Figure 1: Mean sea level pressure (hPa) chart 1200 EST 17/08/98

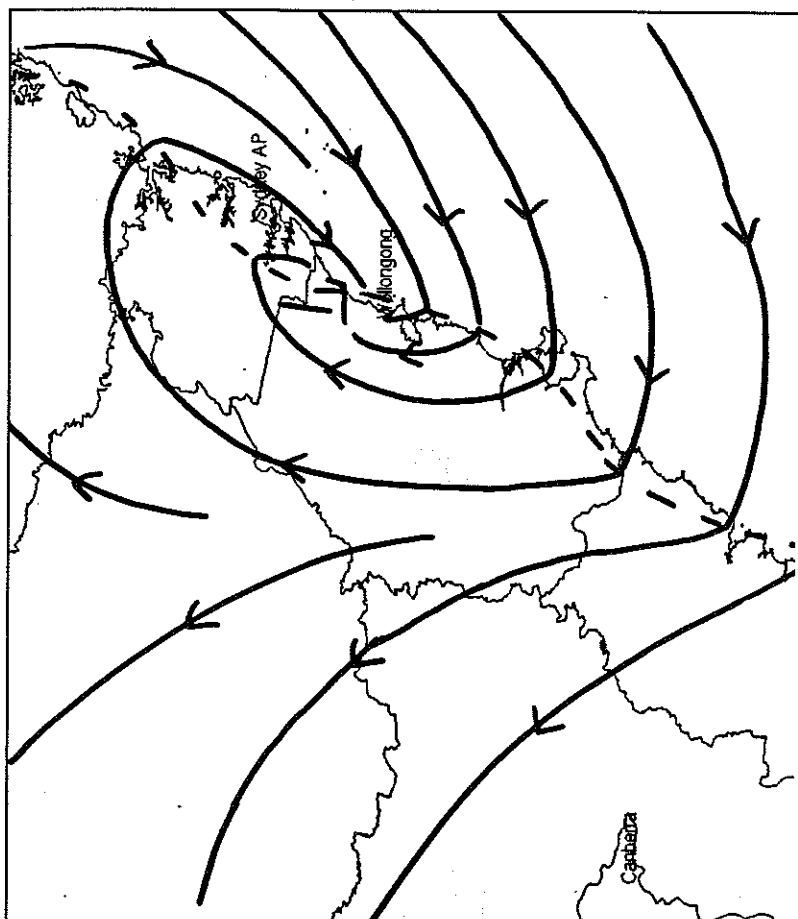


Figure 2: Meso-scale streamline analysis 1800 EST 17/08/98

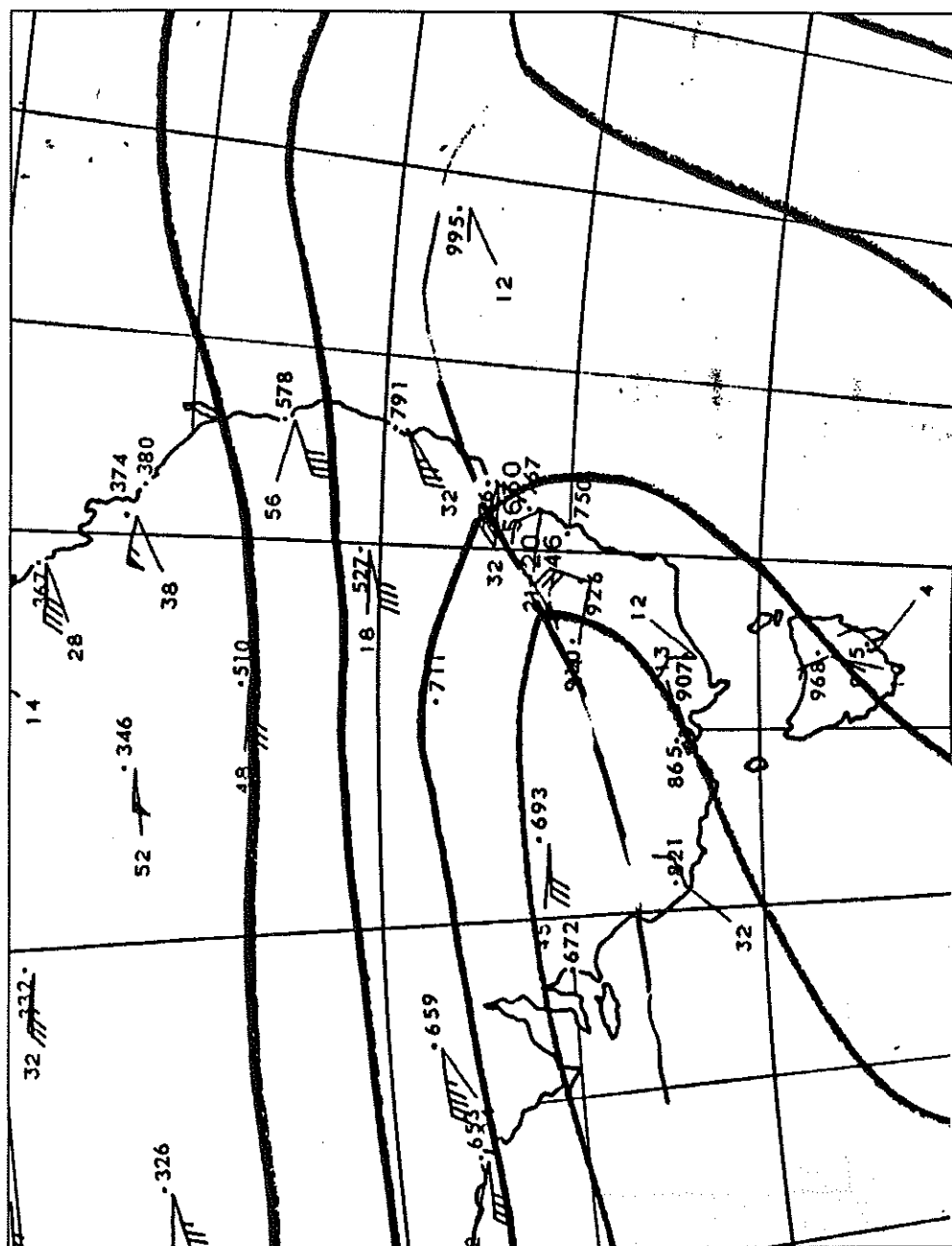


Figure 3: 500hPa Geopotential height (m) analysis, 1500 EST 17/08/98

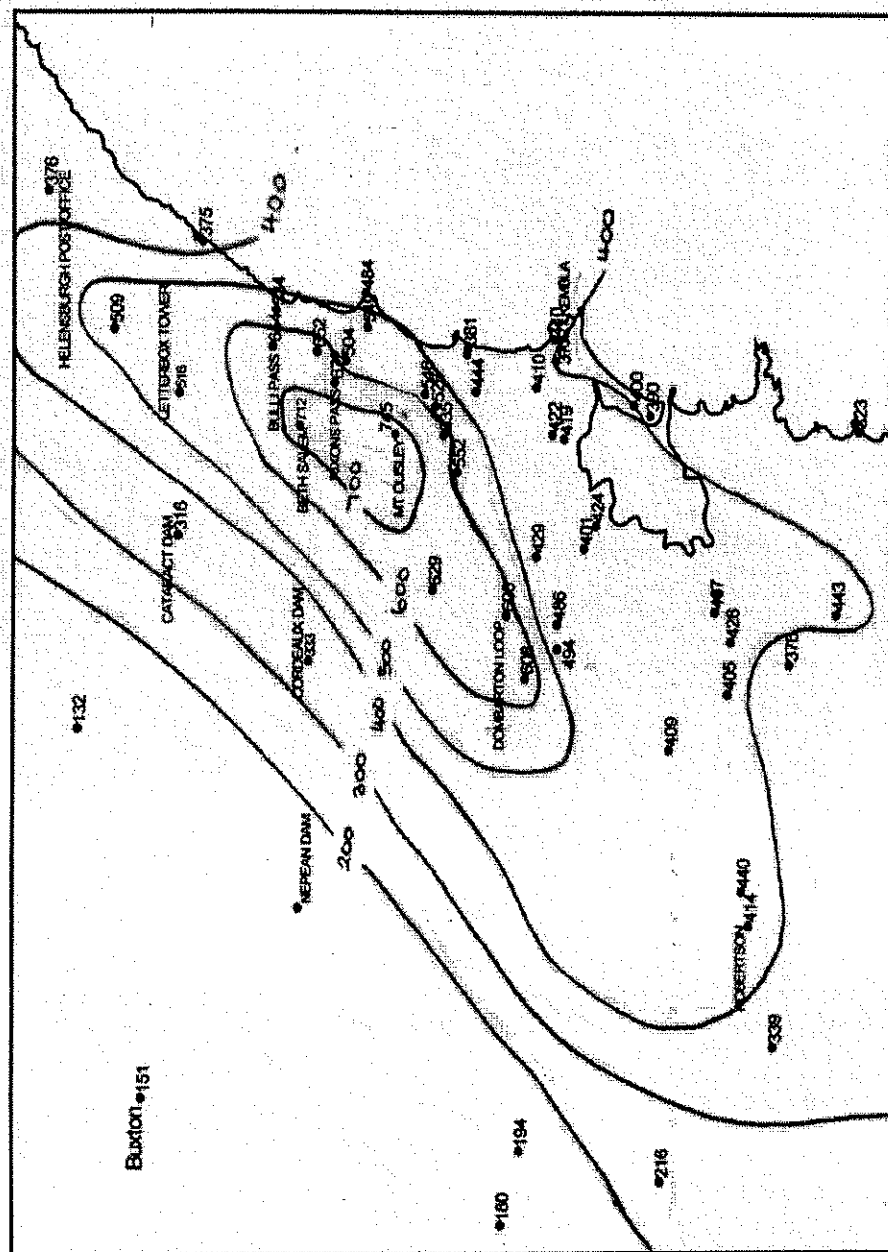


Figure 4: Rainfall totals (mm)
0900 15 August - 0900 20 August 1998

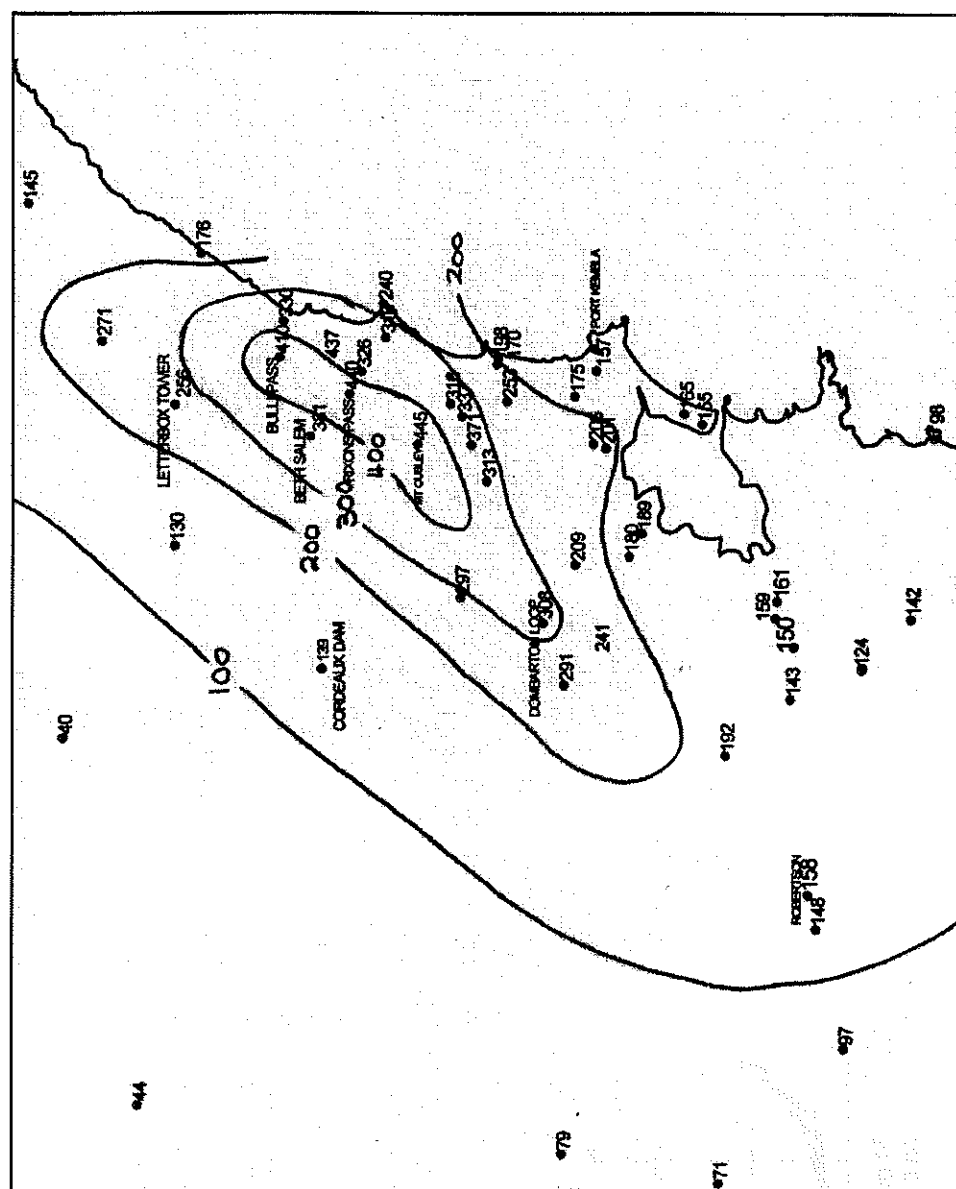


Figure 5: Rainfall totals (mm)
24-hours to 0900 Tuesday 18 August 1998

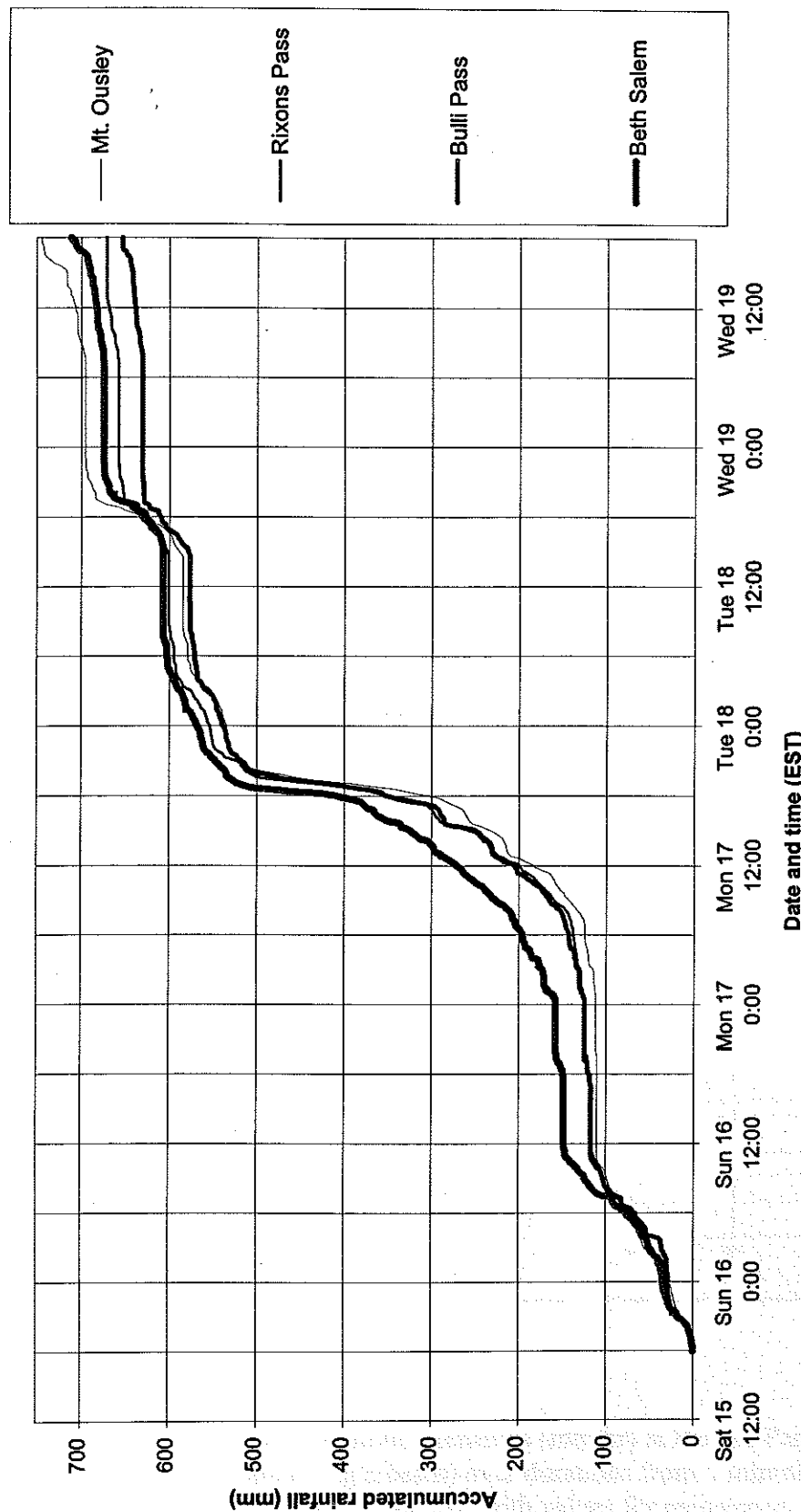


Figure 6: Temporal patterns of accumulated rainfall (mm)
Sat 15 August - Wed 19 August

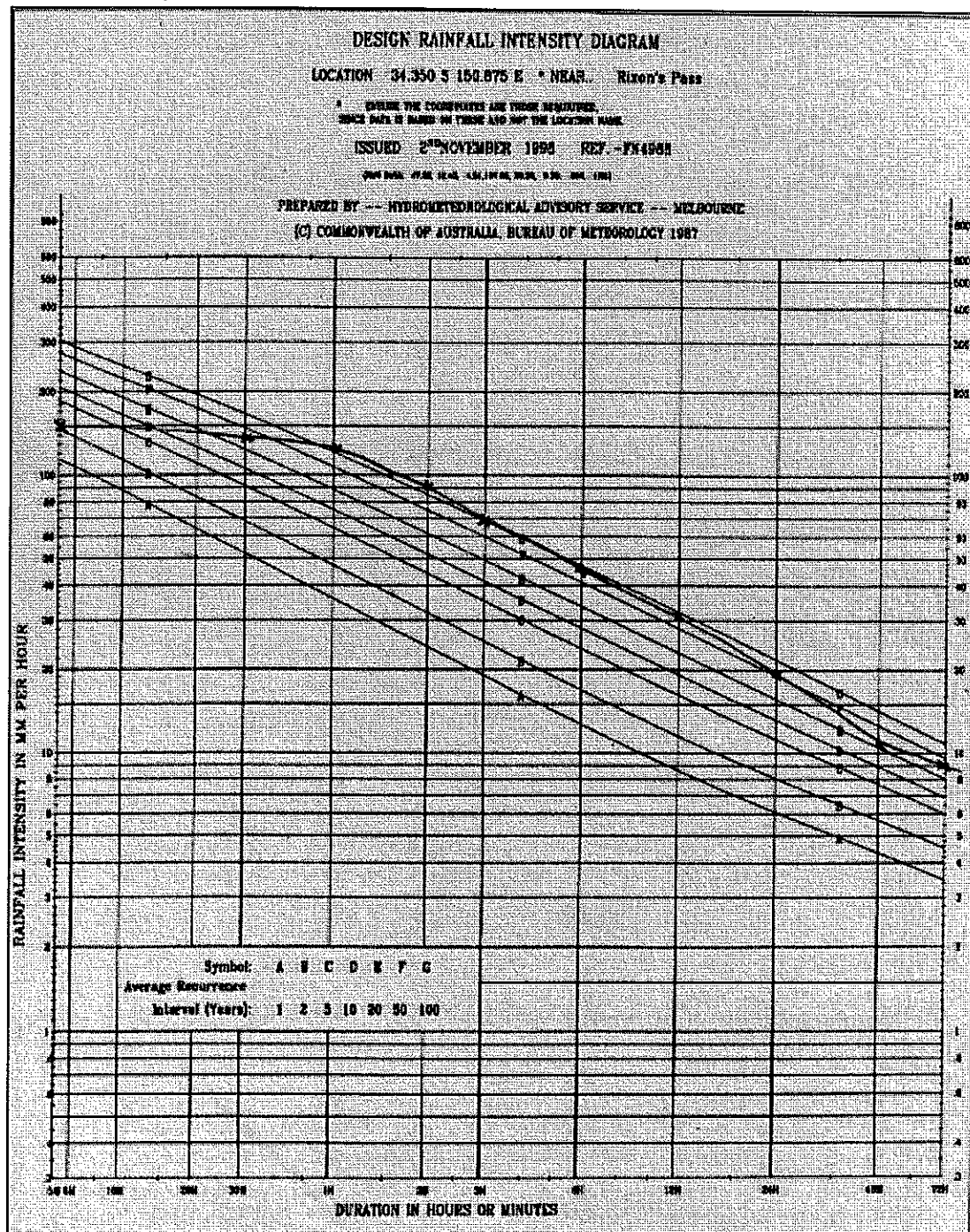


Figure 7: Heaviest rainfall intensities (mm/hr) at Rixons Pass (line joining crosses) over durations from 5 minutes to 72 hours, compared with values for recurrence intervals of 1 year to 100 years.

STATE EMERGENCY SERVICE RESPONSE

Stephen Oppen

State Emergency Service

EMERGENCY RESPONSE TO THE WOLLONGONG FLOODS, AUGUST 1998 – AN SES PERSPECTIVE.

STEVE OPPER, STATE PLANNING COORDINATOR

The event in Wollongong has been described in many different ways but from an SES perspective the event was definitely a flood response operation. In an operational context the SES is not concerned with why water is where it is.

A flood can be considered as water in places where it is not normally supposed to be and which threatens loss of life and/or damage to property. Whether the flood producing rain originates from a heavy downpour directly over the affected area or whether it falls over a distant part of the catchment before flowing through the area, it is only the timing of the SES response that is changed. A flood can also originate from the so called sunny day failure of a water storage structure or the effect of elevated ocean levels during coastal storms.

It is certainly recognised by this Service that not all places affected in Wollongong were inundated by water breaking the banks of watercourses. The heavy rainfall created cross-country flow which clearly caused water to enter buildings which were not near watercourses. The author's own property, high on a hillside, was filled to a water depth of over 150mm because overland flow could not get out under neighbouring steel fences. Within several hundred metres downhill of this property however, it was obvious that watercourses were overflowing and floodwater was rising back along adjacent streets and inundating vehicles, businesses and houses.

From an operational viewpoint the timing of the event could not have been worse. Roads began to close early in the afternoon when many people were trying to get home from work and parents were out driving children to and from activities. When the peak of the heavy rainfall and flooding occurred, it was dark and people could not see what was happening around them.

Early on Monday because of persistent heavy rain, there were indications that serious flooding could occur in the Wollongong area and surrounding suburbs. In response to those signs, the SES Illawarra South Coast Division issued a media release at 12 noon, warning of the potential for flash flooding.

The nature of the landscape of Wollongong presented an additional difficulty. The area is a narrow coastal strip interrupted by many short watercourses. These originate along the cliffs of the escarpment to the west and flow down steep channels to the wetlands and beaches in the east. The main north-south transport links cross these many watercourses. This landscape effectively created dozens of isolated communities cut off from each other by flooded roads.

Emergency services are located in only a few places along this coastal strip and therefore during the 6 hours when conditions were at their worst, only those people directly beside an emergency service facility could be reached.

Many emergency service personnel could not reach their own headquarters until late on Monday night.

Under the State Emergency and Rescue Management Act (1989) (SERM Act) one of the functions of the NSW State Disaster Plan (DISPLAN) is to identify **the combat agency primarily responsible** to deal with a particular emergency (threat). That agency is chosen because it has the expertise and appropriate resources to deal with the threat. Where no combat agency is identified (eg earthquake, landslip), the Local Emergency Operations Controller (LEOCon) or District Emergency Operations Controller (DEOCon) assumes control. When requested to do so by a combat agency, a LEOCon or DEOCon may provide support.

From about 1900hrs to 2300hrs on Monday evening the event was dominated by rescue situations. The NSW Police are the designated combat agency for rescue under DISPLAN. There was also a considerable threat of landslip along the escarpment. There is no designated combat agency for landslip so under DISPLAN the LEOCon at local government level and the DEOCon at emergency management district level (when more than one local government area is affected) assumed control of such landslip incidents. Inter-woven with all this other activity, a major flood response operation was also required and under DISPLAN the SES is the designated combat agency for floods.

During the first 4-6 hours people calling for help rang any telephone number they could find which looked like it could be an emergency service. For example the SES's State Headquarters, an administrative and coordination centre which is coincidentally located in Wollongong, was inundated with calls from the public seeking help. As a result of the calling pattern, the requests for help and assistance from the public were fragmented across almost all emergency services and in turn across many locations.

While operators within each of the services recorded the requests, they were often unable to offer any indication of the nature or timing of a response. This was due to the volume of the requests coming in, the consequential difficulty in trying to prioritise them (except for the obvious life at risk category), the fact that many locations could not be reached, and uncertainty about what emergency service resources were available.

On Monday night, in consultation with the Wollongong City LEOCon, the SES established a dedicated call taking facility at the Illawarra South Coast Division headquarters (also coincidentally located in Wollongong). This service, for people seeking help with flood damage, was advertised over the local radio stations and was effective in taking some pressure off the 000 system. The centre took 5,000 calls during the event.

The scenario of large numbers of people calling for assistance in a short time period is familiar to the SES because in large scale severe storm impacts, a similar calling pattern occurs. This is why the Service uses a sector management approach to provide assistance in large scale events. Instead of trying to manage a massive paper war, dealing with hundreds or even

thousands of recorded telephone requests, the sector approach uses a simple and effective method. Get teams of personnel and equipment out in the field, find the worst affected areas, and get stuck into the job of helping people.

This approach is not guaranteed to find every person in need but it has been proven to get the clean up and recovery process moving long before the alternative paper based approach. Once the bulk of the work is being attended to, a more detailed reconnaissance and media campaign will locate the isolated cases. Getting emergency services into the field where they are visible and directly approachable, also takes the pressure of the telephone based system. This is because people can communicate directly to emergency service workers and most important, can see that a response is happening.

Early in the morning of Tuesday the 18th, the Director General of the SES visited the Local Emergency Operation Centre (LEOC) and it was agreed that the sector approach would be implemented at first light. The SES was already mobilising large numbers of SES and Rural Fire Service (RFS) volunteers from outside Wollongong. SES volunteers were heavily committed to flood and storm damage operations in the southern suburbs of Sydney. Fortunately the RFS were able to respond over 200 of their volunteers from Sydney and these people arrived in Wollongong early Tuesday morning.

For the next several days the response operation was a repetitive large scale resource management task. Personnel and equipment were the main requirements. Personnel from SES, RFS, Police, and NSW Fire Brigades were brought in from many of the adjacent areas of NSW. On any one day there were up to 450 personnel committed to the flood clean-up operation. A total of 41 SES volunteer units sent teams into Wollongong over the seven days of the operation.

The sector management approach got off to a slow start because of an assumption which proved to be invalid. The original plan was that each sector or group of sectors was allocated to a separate emergency service task force under their own control and command. Of the six defined sectors, the two northern sectors were allocated to the SES, the next three middle sectors to the RFS, and the final southern sector to the NSW Fire Brigades.

The assumption was that each task force would have the sector management approach explained to them as they allocated by the LEOC. Never assume anything!! The fire services, because of their specific response training, carried out reconnaissance to look for life in danger, reported results back to a central point, then waited for instructions and tasking. In the SES sectors, where teams were familiar with the concept of large scale flood/storm response work, reconnaissance resulted in immediate local action and clean-up. Reporting back for the SES teams was a matter of confirmation of **tasking decided within the sector** and most important, requesting equipment for the clean-up operation.

When the Director General of SES discovered that there was some misunderstanding about what was required in the sectors, an SES officer was allocated to each of the RFS sectors. Although tasking was then under the control of SES, command of RFS personnel remained with RFS officers. The Sector being managed by NSW Fire Brigades where landslip was a concern, remained under control of the Wollongong City LEOCon.

Supply of equipment and specialist advice eg engineering was also problematic in the early stages. The RFS task force was sending requests for equipment for their sectors to the Local Emergency Operations Centre (LEOC). Within the LEOC these requests could not be processed rapidly enough because Wollongong City Council had committed most of its' resources. In many cases the requests ended up being re-directed to the SES via our liaison officer in the LEOC.

When it became apparent that the LEOC was having difficulty responding to requests for equipment and services, the SES State Operations Centre began to directly source plant, equipment, and services for SES and RFS sectors. This was essential as work was often at a standstill because of lack of resources requested through other channels. Within hours of the change of procedure, large amounts of heavy plant and equipment were on site and this rapidly accelerated the clean-up operation.

The SES has direct access to the Natural Disaster Relief Arrangements (NDRA) funding. Using an Emergency Order system SES can access **any necessary resources** from public agencies or private companies without a complicated red-tape system. The authority to write orders is delegated to the three operational levels with the Service; SES Local Controllers, SES Division Controllers, and SES State Operations Centre Managers. This delegation structure, to ALL operational levels, creates great flexibility and SPEED.

The dollar value of the delegations increases from local up to state level. This provides a self regulating effect so that major resources which need coordination, cannot be committed locally. Because the system is very flexible and can be used freely it is closely scrutinised by both internal and external financial auditors.

Neither the NSW Police nor the NSW Fire Brigades have access to this system because it is linked to natural disasters. The SES as the combat agency for floods and storms, and the RFS as combat agency for bushfire have access for their respective threats. The emergency order system, properly managed, is a most important tool for the SES operations. The Service has resource registers of major suppliers for frequently used goods and services. It is a standard condition that all suppliers on the system are contactable 24 hours a day. The most used and versatile resources register during events however, is the Yellow Pages.

The Wollongong flood demonstrated that despite civil engineering and urban planning intervention, there will be events which exceed the capability of

affordable physical mitigation measures. There are, no doubt, questions to be considered about the appropriateness of the risk levels that apply to development along watercourses in the Wollongong area. From an emergency management perspective, the sensitivity of the road network to local flooding is a critical issue. It is quite amazing that only one person drowned during the event. That fact by itself may suggest something about the severity of this event and extent to which people and property were exposed to it.

If a truly severe or extreme flood event does occur along the escarpment behind Wollongong, how will emergency service cope? Especially during the first 4-6 hours if access is cut in almost every directions. Floodplain management must continue to recognise the importance of emergency management and must accept that urban planning and design need to provide for the continued functioning of emergency services in extreme events. This consideration may mean designing some services such as critical roads, to a higher standard than might apply for day-to-day use.

In the Hawkesbury-Nepean valley where certain roads have been identified as critical for evacuation during floods, a design standard of 1:300 has been adopted for local flooding conditions. This precedent needs to be considered in many other areas of NSW. In floodplain management, emergency management needs to be a planned resource not a last resort.

REFERENCES

NEW SOUTH WALES GOVERNMENT. *STATE EMERGENCY AND RESCUE MANAGEMENT ACT 1989 No 165.*

NSW STATE EMERGENCY MANAGEMENT COMMITTEE. *NEW SOUTH WALES STATE DISASTER PLAN (DISPLAN).*

Insurance Emergency Scheme

John Cooper

Insurance Emergency Scheme

Sydney Water Engineering Panel Institution of Engineers Australia

Wollongong University Function Centre

Tuesday 23 February, 1999

Presentation by:

John Cooper

Chief Adjuster

for the

Insurance Emergency Scheme

The Insurance Council of Australia is an association of most of the Insurers in Australia and its purpose is to act as a spokes body for Australian Insurers in relation to their dealings with the public, Government and with their various association members.

One of the major roles of the ICA also, is to prepare and develop codes of conduct for the members and to co-ordinate and lobby Government's in relation to insurance matters which clearly affect most people in the community.

The ICA has its national headquarters in Sydney with state branches in each of the state capitals.

With the prevalence of natural disasters experienced by the community in Australia, the ICA some years ago created the Insurance Emergency Scheme (IES).

The IES has a national co-ordinator, together with branches in each of the state capitals. The role of the IES is to respond on behalf of the insurance industry to a disaster.

The role of the IES is to initially determine the overall extent of the disaster and to ensure that sufficient and appropriate insurance facilities are available promptly and that personnel are appropriately briefed and able to operate within a short period of time.

The facilities involved include, ensuring there are sufficient Loss Adjusters on site and also to assist in communications with local authorities and various emergency services and other bodies associated with the disaster to ensure that all matters in relation to insurance are made clear and that the provision of insurance services are provided as smoothly and as quickly as possible.

Where problems arise, the presence of IES personnel in the area can resolve these problems and ensure that the appropriate parties are advised and positive action is taken.

In this regard also the IES personnel establish communication channels with the local population which may be through radio and television broadcasts and through advertisements in the newspapers, as occurred in Wollongong during the recent Wollongong storms and floods.

As abovementioned, one of the roles of the IES is also to ensure that there are sufficient Loss Adjusters in the area to handle the substantial quantity of insurance claims.

Individual Loss Adjusters are instructed by, and act on behalf of, insurance companies and their role is to determine if the claim falls within the scope of the Policy concerned and then to adjust the claim within the scope and limitations of that Policy.

This role is independent of the ICA or the IES, with the latter bodies being available in case problems arise in communications and the like which can then be chased up.

In the course of adjusting the claim, the Loss Adjuster often seeks expert advice from specialists such as Structural Engineers, Hydrologists, Accountants, Computer Experts, etc.

In the Wollongong situation where there was clearly a presence of flooding, as per the general definitions within Insurance Policies, there was a need to have expert advice provided from Water Engineers (Hydrologists) to ascertain whether individual claims resulted from damage caused by rainwater, flood, or a mixture of the two.

It should be noted also that individual insurance companies have their own definitions of flood within their Policy wordings and the Hydrologists' reports, together with that of the Adjuster, would have been used to determine acceptance of liability or otherwise in regard to the above.

In the Wollongong disaster period there were a total of 3,582 domestic householder claims recorded, of which approximately 88% were rainwater and/or storm, and were paid without question and 402 claims, being approximately 12%, were declined.

We have been advised that most of the claims that were declined, as a result of flood, have had special arrangements made by Insurers whereby payments were made on an ex-gratia basis without admission of liability. It should be noted that this is a commercial decision taken by Insurers and not a legal decision as required in the Insurance Policy contract.

We understand that there are a small minority of claims still outstanding.

Overall, with the prompt action of the insurance industry, in particular the Loss Adjusters with assistance from the Hydrologists, the vast majority of claims were paid rapidly and with little difficulty or delay.

Inevitably those who were not paid because their Policy did not respond became aggrieved and this quickly resulted in an active response from the Press and Politicians, as no doubt you would all have noticed.

As above indicated, the insurance industry is made up of individual Insurers with their own operating procedures and Policy wordings. The Policies are continually evolving with regard to wordings and definitions as can be demonstrated by examining the Policy texts over the last several years.

With recent disasters, such as those experienced in Wollongong, Coffs Harbour, Townsville, etc., growing demand for flood cover is being experienced and this cover may well be

provided in due course as part of that evolutionary process. Any such cover that is provided will no doubt come at a cost to both those being provided with that cover or to the whole population or Policyholders who may be asked to subsidise those in flood areas.

From the insurance industry's perspective, the rapid response from all parties in relation to the Wollongong disaster was greatly appreciated and this includes the various rescue services in addition to the Hydrologists and those who provided a rapid and immediate reporting response and facility to Insurers so that they could determine cover for individual areas expeditiously.

In retrospect from both the Wollongong and other storm and/or flood disasters, there is a clear need for accurate flood mapping and historical data to be collated of those and other areas which may be affected in the future.

This information will clearly be of great value and provide information so that the appropriate authorities can open up communication channels to implement appropriate building regulations and procedures to avoid development and construction in areas which may cause potential problems in the future.

No doubt with seminars such as this and the use of such post mortem techniques, hopefully followed by appropriate action, loss of property can be greatly reduced.

**“Influence of the Illawarra Escapement on Long
Duration Design Rainfalls and the Implications for
Floodplain Management”**

**Scott Little & Mark Babister
Webb McKeown & Associates**

The Influence of the Illawarra Escarpment on Long Duration Design Rainfalls and the Implications for Floodplain Management

SCOTT LITTLE Engineer, Webb McKeown & Associates

MARK BABISTER Associate, Webb McKeown & Associates

1 INTRODUCTION

The Illawarra Escarpment is located along the NSW coastline some 70 km south of Sydney. The sandstone cliffs of the escarpment rise steeply from the talus material at the base to form shear faces 200 to 300 metres in height. The escarpment extends through populated areas from Stanwell Park in the north, running close to parallel with the coastline for 90 km to Nowra in the south. The city of Wollongong is situated approximately midway along, and below, the escarpment.

The physical nature of the Illawarra Escarpment, together with its closeness to the coastline, results in unique and very localised meteorological effects. Drainage catchments at the base of the Illawarra Escarpment have been subject to very intense rainfall bursts resulting in flash flooding and triggering landslides. The severity of these storms and concern for the region's considerable infrastructure investment, together with the unique and very localised meteorological effects displayed by the Illawarra Escarpment have justified specific investigations aimed at refining design rainfall estimates produced by Australian Rainfall and Runoff, 1987 (referred to here as AR&R87 (1)) within the Illawarra region.

This paper draws upon the findings of recent undergraduate research and examines the significance of increased amounts of available rainfall data on long duration rainfall estimates in the Illawarra region. Furthermore, the impact of the quality of rainfall data on long duration design rainfalls is detailed.

2 AVAILABLE RAINFALL DATA

An extensive array of rainfall data are available for the Illawarra region, with a number of organisations actively participating in rainfall data collection. Since the publication of AR&R87 (1), more than ten years of rainfall data has been collected at existing gauges, with many more organisations installing rainfall recording gauges in response to a number of significant storm events. This progressive installation of rainfall gauges in the Illawarra region over the past ten to fifteen years has made the local rain gauge network more dense.

Organisations including the Bureau of Meteorology (BoM), Australian Water Technologies (AWT) and Manly Hydraulics Laboratory (MHL) operate rainfall recording gauges in the Illawarra region. Over one hundred current and non current BoM rainfall gauges are identified within 50 km of Wollongong City. Long lengths of record are obtainable from BoM daily-read rainfall records, some of which contain over 100 years of recorded data. AWT have records from some 60 current and non current rainfall gauges within the same area. Daily-read rainfall records are of a similar length to that of the BoM records. MHL operates approximately 17 current pluviograph rainfall gauges within 25 km of Wollongong City. Pluviograph gauges operated by MHL were first installed in the early 1980's and available lengths of record are therefore much shorter than the daily-read data available from many of the BoM and AWT records.

As shown in Figure 1, the network of BoM stations indicates a good broad coverage below, above and greater than 20 km inland of the escarpment. A lack of data is evident immediately above and below the escarpment face, and in the catchment areas for several of the Sydney and Wollongong water supply dams. The network of AWT stations, although not as dense as that of the BoM network provides a good coverage within the area for which BoM data is deficient. The pluviograph rainfall gauges operated by MHL are concentrated around

the built up areas of Wollongong and surrounding suburbs. A number of these gauges are located directly above or below the Illawarra Escarpment.

Other organisations in possession of rainfall data in the Illawarra region include the Rail Services Authority (RSA), the Roads and Traffic Authority (RTA), Illawarra Coke Company and the University of NSW. These organisations typically maintain a small number of rainfall gauges with a short length of record.

3 DESIGN RAINFALL ESTIMATION

3.1 Australian Rainfall & Runoff

The current method of design rainfall estimation for any location in Australia employs algebraic or graphical procedures detailed in AR&R87 (1). These procedures produce design Intensity-Frequency-Duration (IFD) curves for durations from 5 minutes to 72 hours and frequencies up to the 100 year average recurrence interval (ARI) and enables extrapolation of design rainfall estimates to a 500 year ARI.

The data used to establish the method of design rainfall estimation presented in AR&R87 (1) was extracted from the BoM Australia wide rain gauge network. The results of the analysis from the BoM rainfall data are presented as six master charts in AR&R87 (1) Volume 2. These master charts cover all regions of Australia and contain isopleths of log-normal design rainfall intensities of 1, 12 and 72 hour durations for 2 year and 50 year ARIs. Also contained in AR&R87 (1) Volume 2 is a regionalised map of skewness and two maps containing geographical factors, used to obtain design rainfall estimates for durations less than 1 hour. The purpose of these maps is to provide a regional data set from which design rainfall intensities can be derived by interpolation or extrapolation to a range of durations and ARIs.

3.2 Recent Research

An undergraduate research thesis completed by Kennedy and Little (2) in July 1998 investigated the influence of the Illawarra Escarpment on long duration design rainfall estimates. The investigation made use of daily-read rainfall records only. Pluviograph data were not made available due to the high cost associated with extraction of this data. As such, the conclusions drawn from this investigation are applicable only to design rainfall estimates for durations of 24 hours or greater.

The investigation involved the analysis of large quantities of BoM and AWT daily-read rainfall records in contrast to the analysis undertaken for AR&R87 (1), which primarily involved only BoM rainfall records. BoM and AWT daily-read rainfall records were obtained from some 150 current and non-current rainfall gauges within 50 km of Wollongong City, of which approximately 35 had a continuous length of record greater than 30 years. As illustrated in Figure 1, incorporating AWT rainfall data, together with longer lengths of record resulted in a more comprehensive rainfall gauge network than available for past studies, particularly immediately above and below the escarpment face.

Detailed data checking for homogeneity, completeness and reliability, in conjunction with the disaggregation of accumulated data resulted in the extraction of 24 hour and 72 hour annual maxima data sets. At-site log-normal frequency analysis of the extracted data sets, incorporating the conversion of restricted to unrestricted rainfall totals and converting probabilities of occurrence from AEPs (obtained from frequency analysis of an annual data series) to ARIs enabled direct comparison of rainfall estimates to those produced in AR&R87 (1). This comparison indicated an increase in design rainfall estimates of up to 15% in areas immediately above and below the escarpment face.

The areas of greatest increase coincide with areas in which the BoM rainfall gauge network is sparse, being immediately above and below the escarpment face. Areas of negligible change are observed more toward the coast and inland of the escarpment, coinciding with areas in which the BoM gauge network has a good broad coverage. It is shown that current long duration design rainfall estimates produced from AR&R87 (1) in the vicinity of the Illawarra Escarpment are based on data obtained from a spatially inadequate rainfall gauge network. This spatial inadequacy is due to a lack of rainfall gauges in areas subject to high rainfall volumes, being immediately above and below the escarpment face. Therefore, current long duration design rainfalls are based on unrepresentative daily-read rainfall records.

4 DATA QUALITY

Poor quality rainfall data has the potential to adversely influence design rainfall estimates. Further findings to be drawn from analysis of the extracted annual maxima data sets relates to the potential for daily-read gauge overflows in high rainfall areas and subsidiary impact on regional skewness.

4.1 Gauge Overflows

From frequency plots of annual maxima data sets, certain records appeared to be "capped" at a rainfall depth of approximately 300 mm as shown in Figure 2. This capping effect is thought to be a product of the inclusion of artificially low rainfall depths resulting from gauge overflows. Gauge overflows may have occurred in large storms due to an insufficient capacity of daily-read gauges used in the area. Adding further weight to this argument is the observation that data sets containing these suspected gauge overflows were predominately recorded close to the escarpment where storm intensities are greatest. The inclusion of spurious low annual maxima reduces the sample mean and standard deviation of the data set and therefore further reduces current design rainfall estimates.

4.2 Regional Skewness

As shown in Figure 2, the inclusion of artificially low annual maxima data lowers the sample skewness, commonly taking on a negative value. Australia wide regional skewness coefficients presented in AR&R87 (1) Volume 2 have a lower bound of zero and an upper limiting value of 0.8. Therefore, when analysis of Australia wide BoM rainfall records was undertaken for AR&R87 (1), negative skewness, if encountered was set to zero.

From mapping of at-site skewness, zones of predominately positive skewness were observed toward the coast and some 10 km to 20 km inland of the escarpment. As illustrated in Figure 3, a large zone of predominately negative skewness was identified in around the Illawarra Escarpment. It is reasonable to suggest this large zone of negative skewness resulted in AR&R87 (1) Volume 2 presenting a regional skewness of zero for the Illawarra region. It is believed the area of negative skewness around the Illawarra Escarpment is a result of gauge overflows, and as such, regional skewness is more likely to be in excess of zero. The implication of a non-zero, positive regional skewness would be to change current design rainfall estimates from log-normal to log Pearson Type III therefore further increase long duration design rainfall estimates for low frequency events.

5 IMPLICATIONS FOR FLOODPLAIN MANAGEMENT

Following the recent flash flood event on 17 August 1998 it has been proposed to re-evaluate current flood studies and floodplain management plans for drainage catchments in the Illawarra region. The findings detailed in this paper highlight the need for a revision of current design rainfall estimates in conjunction with any re-evaluation of flood studies and floodplain management plans.

Although the findings detailed within this paper pose a serious threat to the current design standard in the Illawarra region, it must be noted that the conclusions are relevant only to long duration design rainfall estimates. Drainage catchments within the Illawarra region have a

time of concentration well below that of 24 hours and as such, the need exists for the investigation to be extended to include the analysis of short duration pluviograph data.

Furthermore, the ramifications of this recent research may not be isolated to floodplain management issues within the Illawarra region alone. Design rainfalls in other coastal New South Wales areas influenced to some degree by topography may also exhibit similar spatial and data quality problems as those encountered in the Illawarra region.

6 CONCLUSIONS

Current long duration design rainfall estimates produced by AR&R87 (1) may be in error by up to 15% in the vicinity of the Illawarra Escarpment. This discrepancy is not due to the method of design rainfall estimation itself, but rather the data utilised in establishing the method. Increases in long duration design rainfall estimates are the result of utilising data recorded from a more comprehensive rainfall gauge network than available for past studies. This has the impact of enabling design rainfall estimates to be based on data thought to be more representative of rainfall depths received in the vicinity of the Illawarra Escarpment.

Furthermore, it is concluded that annual maxima rainfall depths extracted from daily-read rainfall records may include artificially low recorded depths due to the possible occurrence of gauge overflows. This has a significant impact on the average regional skewness coefficient for the Illawarra region.

7 REFERENCES

1. I.E.Aust. 1987. *Australian Rainfall And Runoff*. Ed. Pilgrim, D.H., Volumes 1 & 2, (Institution of Engineers: Australia)
2. Kennedy and Little. 1998. *The Influence of the Illawarra Escarpment of Design Rainfalls*. Undergraduate thesis, University of Technology, Sydney.

FIGURE 1

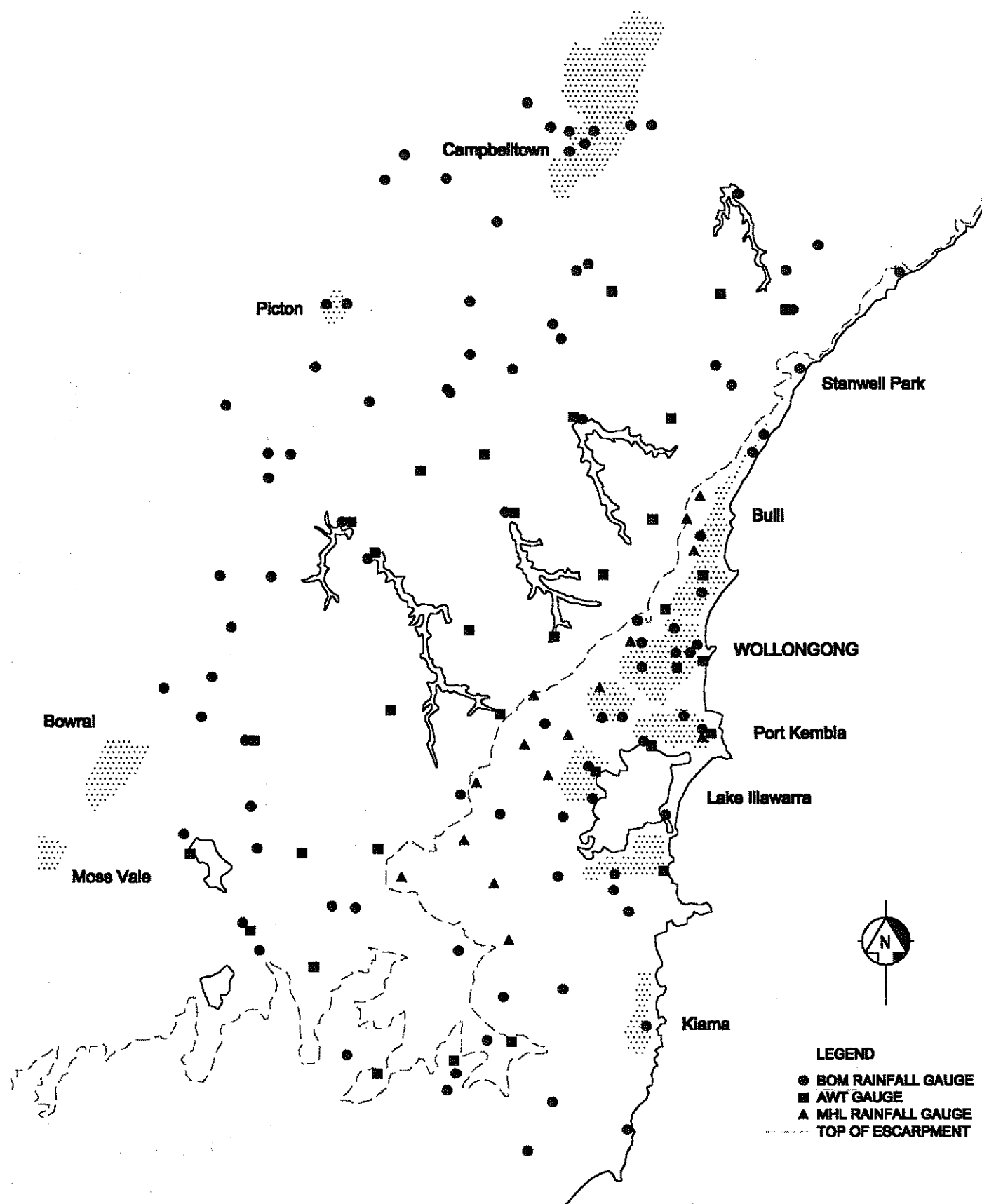
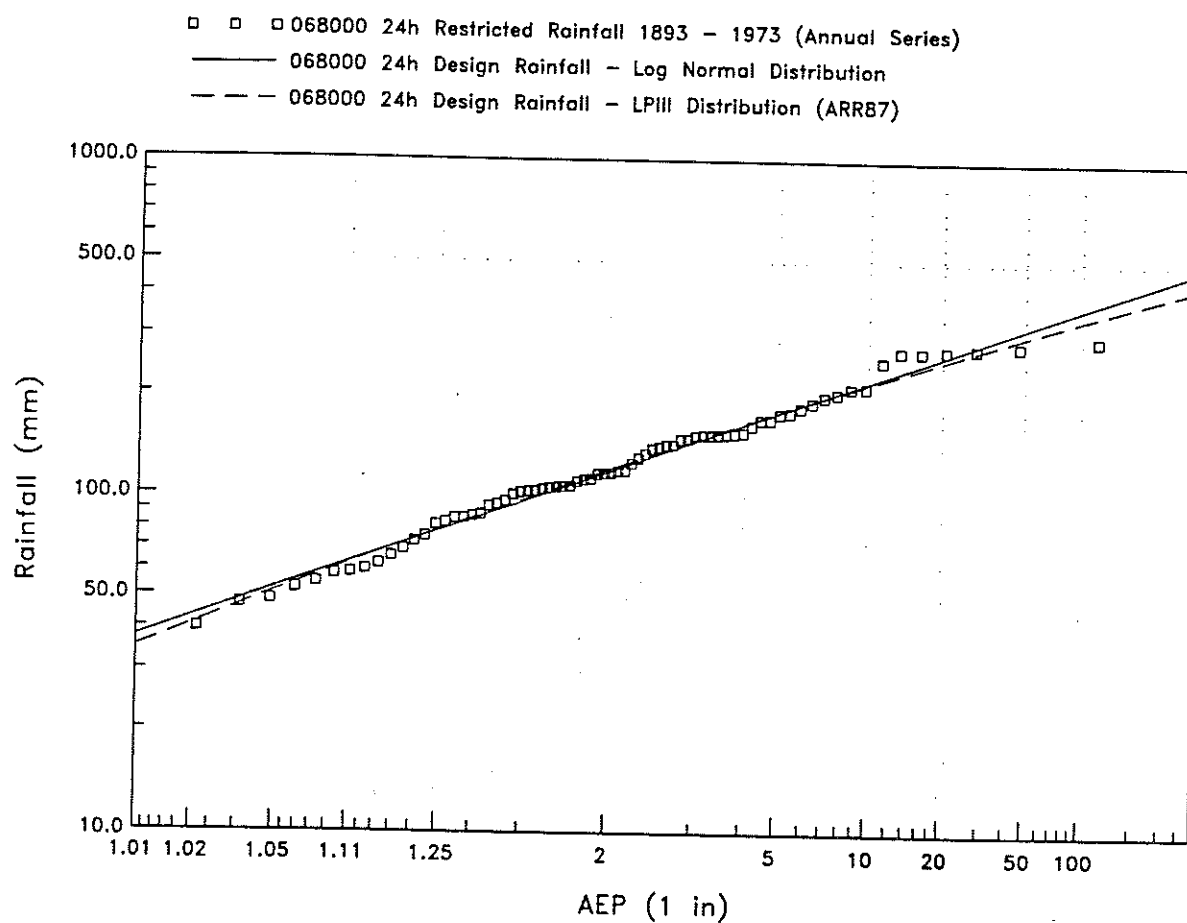


FIGURE 2



18/07/98 14:24

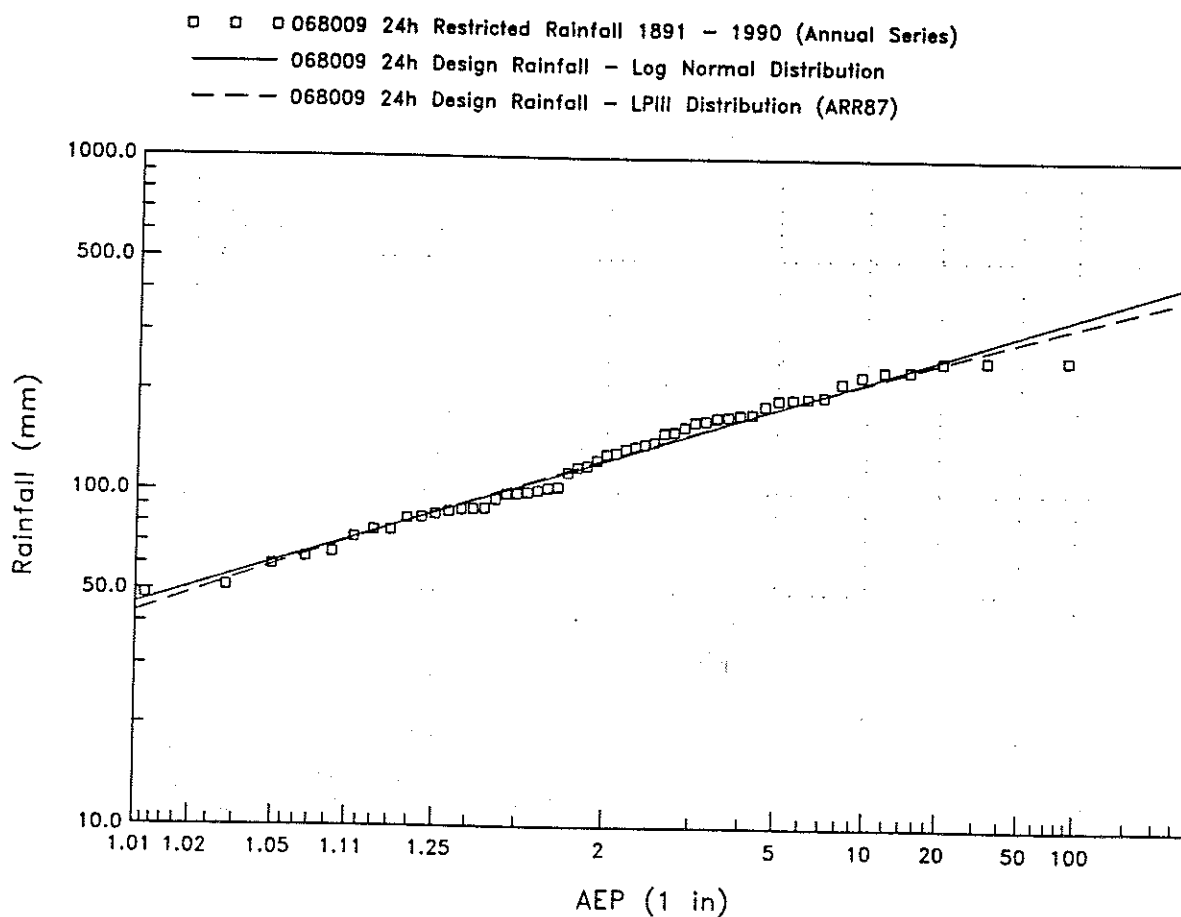


FIGURE 3

